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SECTION 1: TUBE SKIN TEMPERATURE LIMITS & CREEP PRESERVATION

1.1 Maximum Allowable Tube Skin Temperatures (API 530 Design Margins):

- Carbon Steel (A106-B): Max 454 deg C.
- 5% Cr - 0.5% Mo (ASTM A335 P5): Max 593 deg C.
- 9% Cr - 1% Mo (ASTM A335 P9): Normal limit 650 deg C. Emergency Alarm at 685 deg C.
- Operation above 700 deg C drastically reduces creep life by 90% per 15 deg C increment.

SECTION 2: FIREBOX DRAFT & COMBUSTION CONTROL

2.1 Arch Draft: Must strictly be maintained negative between -2.0 and -5.0 mm H2O.

2.2 Excess O2 Target: 2.5% to 3.5% at arch. High O2 accelerates outer tube scale formation.

SECTION 3: STEAM-AIR DECOKING CRITERIA & PIGGING

3.1 Tube Decoking Trigger: When clean tube skin vs process fluid temperature Delta-T > 95 deg C.

3.2 Mechanical Decoking (Pigging): Required when pressure drop across radiant pass increases > 35%.

3.3 Spalling Verification: Monitor effluent quench water until zero carbon fines are observed.